Chapter 4. Nonstationary Time Series Models and Model Selection

ARIMA, unit root tests, model selection and diagnostics

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4.1 Autoregressive integrated moving average models

Motivation. Many observed series (prices, output, consumption) wander without returning to a fixed level, so the stationary ARMA theory of the previous chapters does not apply directly. The simplest remedy is to difference the series until it becomes stationary and then fit an ARMA model to the differences; this gives the ARIMA class.

Consider the AR(1) recursion with coefficient one,

$$X_t = X_{t-1} + \epsilon_t, \qquad \text{equivalently} \qquad X_t = X_0 + \epsilon_1 + \cdots + \epsilon_t,$$

the *random walk*. With the backshift operator B it reads $(1-B)X_t = \epsilon_t$, i.e. $\nabla X_t = \epsilon_t$ with $\nabla = 1 - B$.

Example (the random walk is not stationary)

Let $X_0 = 0$ and $\{\epsilon_t\} \sim \text{IID}(0, \sigma^2)$. Then $EX_t = 0$ but $\text{Var}(X_t) = t\sigma^2$ and $\text{Cov}(X_t, X_{t+h}) = t\sigma^2$ for $h \geq 0$, which depend on t; hence $\{X_t\}$ is not stationary. Its first difference $\nabla X_t = \epsilon_t$ is white noise. Equivalently, the AR polynomial $\phi(z) = 1 - z$ has its root on the unit circle, so the stationarity condition of Chapter 2 fails.

In general a nonstationary series may become stationary after $d \geq 1$ differences: $Z_t = (1-B)^d X_t$ is stationary. If moreover Z_t is an ARMA process, $\phi(B)Z_t = \theta(B)\epsilon_t$, then the model for $\{X_t\}$ is $\phi(B)(1-B)^d X_t = \theta(B)\epsilon_t$.

Definition 4.1.1 (ARIMA(p,d,q) model)

A time series $\{X_t\}$ follows an *autoregressive integrated moving average model* of orders p, d, q, written $\{X_t\} \sim \text{ARIMA}(p,d,q)$, if

$$\phi(B)(1-B)^d X_t = \theta(B)\epsilon_t, \qquad \{\epsilon_t\} \sim \text{IID}(0, \sigma^2),$$

where $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$ and $\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$. In practice $d \in \{0, 1, 2\}$.

Throughout, $\phi(z)$ and $\theta(z)$ are assumed to satisfy the causality and invertibility conditions (no zeros in $|z| \leq 1$), so that the d-th difference is a causal invertible ARMA process. The errors are taken IID rather than merely white noise because the inference below (limit distributions, likelihood) needs it.

Remark 4.1.2

The random walk is $\text{ARIMA}(0,1,0)$. If the differenced series has a nonzero mean μ, the model is written

$$\phi(B)\big((1-B)^d X_t - \mu\big) = \theta(B)\epsilon_t, \qquad \mu = E(1-B)^d X_t.$$

Polynomial trends. Not every nonstationary series becomes stationary by differencing, but series with polynomial trend do. For $X_t = t + \epsilon_t$ with $\{\epsilon_t\} \sim \text{IID}(0, \sigma^2)$,

$$(1-B)X_t = t + \epsilon_t - (t-1) - \epsilon_{t-1} = 1 + \epsilon_t - \epsilon_{t-1},$$

so $(1-B)X_t - 1 = \theta(B)\epsilon_t$ with $\theta(B) = 1 - B$: $\{X_t\}$ is $\text{ARIMA}(0,1,1)$ with $\mu = 1$ (with a non-invertible MA part). More generally, for $X_t = a_0 + a_1 t + \cdots + a_r t^r + \epsilon_t$ the series $(1-B)^r X_t$ is stationary: $(1-B)$ maps a polynomial in t of degree $k \geq 1$ to one of degree $k-1$ (the leading term at^k becomes $a\{t^k - (t-1)^k\} = ak\,t^{k-1} + \cdots$), so $(1-B)^r$ turns the trend into the constant $a_r r!$, while

$(1-B)^r\epsilon_t$ is a finite moving average of IID noise. Thus polynomial-trend models can be rewritten as ARIMA models. The converse is false, and when a series is known to follow a polynomial trend it is preferable to fit that trend directly.

4.2 Unit root tests

Motivation. Whether to difference is itself a statistical question: does the AR polynomial have a root at $z = 1$? Under a unit root the least-squares estimator converges at rate n rather than $\sqrt{n}$ and its limit is a functional of Brownian motion, not a normal law, so special (Dickey–Fuller) critical values are needed.

4.2.1 The AR(1) case

Consider $X_t = \phi X_{t-1} + \epsilon_t$, $X_0 = 0$, $\{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2)$, and the hypotheses

$$H_0 : \phi = 1 \qquad \text{vs} \qquad H_1 : |\phi| < 1.$$

Let $\hat{\phi}_{LS} = \sum_{t=1}^{n} X_{t-1}X_t / \sum_{t=1}^{n} X_{t-1}^2$ be the least-squares estimator and

$$T_n = n(\hat{\phi}_{LS} - 1). \tag{4.1}$$

Under H_0, $X_t - X_{t-1} = \epsilon_t$, hence $\hat{\phi}_{LS} - 1 = \sum X_{t-1}\epsilon_t / \sum X_{t-1}^2$ and

$$T_n = \frac{n^{-1}\sum_{t=1}^{n} X_{t-1}\epsilon_t}{n^{-2}\sum_{t=1}^{n} X_{t-1}^2}. \tag{4.2}$$

The scaling suggests looking at the partial-sum process

$$W_n(s) = \frac{1}{\sqrt{n}\,\sigma} \sum_{j=1}^{[ns]} \epsilon_j, \qquad 0 \le s \le 1, \tag{4.3}$$

where $[\cdot]$ is the integer part. Under H_0, $X_t = \sigma\sqrt{n}\, W_n(t/n)$.

Definition 4.2.1 (Brownian motion)

A stochastic process $\{W(s) : s \ge 0\}$ with continuous sample paths is a (standard) *Brownian motion* if
(1) $W(0) = 0$;
(2) for all $t > s \ge 0$, $W(t) - W(s) \sim N(0, t-s)$;
(3) for all $0 \le t_1 < \cdots < t_k$, the increments $W(t_1)$, $W(t_2) - W(t_1), \ldots, W(t_k) - W(t_{k-1})$ are independent.

The convergence below is weak convergence of random elements of the space $D[0,1]$ of right-continuous functions with left limits, equipped with the Skorokhod topology: $W_n \xrightarrow{d} W$ means $Ef(W_n) \to Ef(W)$ for every bounded continuous $f : D[0,1] \to \mathbb{R}$. By the continuous mapping theorem, then $h(W_n) \xrightarrow{d} h(W)$ for every measurable h that is continuous at almost every path of W.

Theorem 4.2.2 (Functional central limit theorem)

Let $\{\epsilon_t\} \sim \mathrm{IID}(0, \sigma^2)$ and define W_n by (4.3). Then, as $n \to \infty$, $W_n \xrightarrow{d} W$, a standard

Brownian motion on $[0,1]$.

Proof. [Supplement] Write $S_k = \epsilon_1 + \cdots + \epsilon_k$, so $W_n(s) = S_{[ns]}/(\sigma\sqrt{n})$. Weak convergence in $D[0,1]$ follows from (a) convergence of finite-dimensional distributions and (b) tightness of $\{W_n\}$ (Prokhorov's theorem, used as a tool); we verify both.

Step 1 (finite-dimensional laws). Fix $0 = s_0 < s_1 < \cdots < s_k \le 1$ and put $m_j = [ns_j] - [ns_{j-1}]$, so $m_j/n \to s_j - s_{j-1} > 0$. The increment

$$W_n(s_j) - W_n(s_{j-1}) = \sqrt{\frac{m_j}{n}} \cdot \frac{1}{\sigma\sqrt{m_j}} \sum_{i=[ns_{j-1}]+1}^{[ns_j]} \epsilon_i$$

converges in distribution to $N(0, s_j - s_{j-1})$ by the classical CLT and Slutsky's lemma. The k increments are built from disjoint blocks of the ϵ_i, hence independent, so their joint characteristic function factorizes and the vector of increments converges to a vector of independent $N(0, s_j - s_{j-1})$ variables, which is the law of the increments of W. Applying the linear map (increments)$\mapsto$(partial sums) gives $(W_n(s_1), \ldots, W_n(s_k)) \xrightarrow{d} (W(s_1), \ldots, W(s_k))$.

Step 2 (a maximal inequality). For partial-sum processes of IID variables a sufficient condition for tightness (Billingsley, *Convergence of Probability Measures*, Thm. 8.4, used as a tool) is: for every $\varepsilon > 0$ there are $\lambda > 1$ and n_0 with

$$P\Big(\max_{k \le n} |S_k| \ge \lambda\sigma\sqrt{n}\Big) \le \frac{\varepsilon}{\lambda^2} \qquad (n \ge n_0).$$

By Etemadi's inequality, $P(\max_{k\le n}|S_k| \ge 3a) \le 3\max_{k\le n} P(|S_k| \ge a)$. Take $a = \mu\sigma\sqrt{n}$ with $\mu = \lambda/3$. Let $N \sim N(0,1)$. By the CLT there is k_μ such that $P(|S_k| \ge \mu\sigma\sqrt{k}) \le 2P(|N| \ge \mu)$ for $k \ge k_\mu$, and $2P(|N| \ge \mu) \le 2EN^4/\mu^4 = 6/\mu^4$. Hence

(i) for $k_\mu \le k \le n$: $P(|S_k| \ge \mu\sigma\sqrt{n}) \le P(|S_k| \ge \mu\sigma\sqrt{k}) \le 6/\mu^4$;

(ii) for $k < k_\mu$: by Chebyshev, $P(|S_k| \ge \mu\sigma\sqrt{n}) \le k/(\mu^2 n) \le k_\mu/(\mu^2 n) \le 1/\mu^4$ once $n \ge \mu^2 k_\mu$.

Therefore $P(\max_{k\le n}|S_k| \ge \lambda\sigma\sqrt{n}) \le 3 \cdot 7/\mu^4 = 1701/\lambda^4$ for large n, which is $\le \varepsilon/\lambda^2$ as soon as $\lambda^2 \ge 1701/\varepsilon$.

Step 3 (conclusion). By Steps 1–2 every subsequence of $\{W_n\}$ has a further subsequence converging weakly, and every limit has the finite-dimensional laws of Brownian motion; since these determine the law on $D[0,1]$ (and Brownian motion has continuous paths), the whole sequence converges to W. ∎

Proposition (Limit of the unit-root statistics)

Under $H_0 : \phi = 1$, with T_n as in (4.1),

$$T_n \xrightarrow{d} T := \frac{\frac{1}{2}(W(1)^2 - 1)}{\int_0^1 W(s)^2\,ds} = \frac{\int_0^1 W(s)\,dW(s)}{\int_0^1 W(s)^2\,ds}.$$

Under $H_1 : |\phi| < 1$, $P(|T_n| \ge c) \to 1$ for every constant c; that is, the test rejecting for large $|T_n|$ is consistent.

Proof. [Supplement]

Step 1 (numerator). Squaring $X_t = X_{t-1} + \epsilon_t$ gives $X_t^2 - X_{t-1}^2 = 2X_{t-1}\epsilon_t + \epsilon_t^2$. Summing over $t = 1, \ldots, n$ with $X_0 = 0$,

$$\frac{1}{n}\sum_{t=1}^{n} X_{t-1}\epsilon_t = \frac{1}{2}\Big(\frac{X_n^2}{n} - \frac{1}{n}\sum_{t=1}^{n}\epsilon_t^2\Big) = \frac{\sigma^2}{2}\Big(W_n(1)^2 - \frac{1}{n\sigma^2}\sum_{t=1}^{n}\epsilon_t^2\Big).$$

Step 2 (denominator). For $s \in [(t-1)/n, t/n)$ we have $[ns] = t-1$, so $W_n(s) = X_{t-1}/(\sigma\sqrt{n})$. Hence exactly

$$\frac{1}{n^2}\sum_{t=1}^{n} X_{t-1}^2 = \sigma^2 \sum_{t=1}^{n} \int_{(t-1)/n}^{t/n} W_n(s)^2\, ds = \sigma^2 \int_0^1 W_n(s)^2\, ds.$$

Step 3 (joint limit). The map $w \mapsto (w(1), \int_0^1 w(s)^2 ds)$ from $D[0,1]$ to $\mathbb{R}^2$ is continuous at every continuous w (Skorokhod convergence to a continuous limit is uniform convergence). By Theorem 4.2.2 and the continuous mapping theorem, $(W_n(1), \int_0^1 W_n^2) \xrightarrow{d} (W(1), \int_0^1 W^2)$; by the law of large numbers $n^{-1}\sum \epsilon_t^2 \xrightarrow{p} \sigma^2$, and Slutsky's lemma gives joint convergence of all three quantities.

Step 4 (ratio). Since $\int_0^1 W(s)^2 ds > 0$ almost surely, the map $(a,b,c) \mapsto \frac{1}{2}(a^2 - c)/b$ is continuous at the limit, and Steps 1–3 yield $T_n \xrightarrow{d} \frac{1}{2}(W(1)^2 - 1)/\int_0^1 W^2$. The second form follows from Itô's formula $W(1)^2 = 2\int_0^1 W\, dW + 1$.

Step 5 (consistency). If $|\phi| < 1$, write $T_n = n(\hat\phi_{LS} - \phi) + n(\phi - 1)$. By the asymptotic normality of the least-squares estimator for a causal AR(1) (Chapter 3), $\sqrt{n}(\hat\phi_{LS} - \phi) \xrightarrow{d} N(0, 1-\phi^2)$, so $n(\hat\phi_{LS} - \phi) = O_p(\sqrt{n})$, whereas $n(\phi - 1) \to -\infty$ linearly. Thus $T_n/n \xrightarrow{p} \phi - 1 < 0$ and $P(|T_n| \geq c) \to 1$. ∎

Notice the contrast with the stationary case: the estimator converges at rate n (super-consistency) and the limit T is not normal (it is skewed to the left). The test rejects H_0 at level α when $|T_n| > c_\alpha$, where $P(|T| > c_\alpha) = \alpha$; the quantiles of T are tabulated by simulation.

Dickey–Fuller test. The studentized version is

$$DF_n = \Big(\frac{\sum_{t=1}^n X_{t-1}^2}{\hat\sigma^2}\Big)^{1/2}(\hat\phi_{LS} - 1), \qquad \hat\sigma^2 = \frac{1}{n}\sum_{t=1}^{n}(X_t - \hat\phi_{LS} X_{t-1})^2.$$

Under H_0,

$$DF_n = \frac{n^{-1}\sum_{t=1}^n X_{t-1}\epsilon_t}{\hat\sigma\,(n^{-2}\sum_{t=1}^n X_{t-1}^2)^{1/2}} \xrightarrow{d} DF := \frac{\frac{1}{2}(W(1)^2 - 1)}{\big(\int_0^1 W(s)^2 ds\big)^{1/2}}.$$

Indeed, by Steps 1–3 above the numerator and the bracket in the denominator converge jointly to $\frac{\sigma^2}{2}(W(1)^2 - 1)$ and $\sigma^2\int_0^1 W^2$, and $\hat\sigma^2 \xrightarrow{p} \sigma^2$: expanding $X_t - \hat\phi_{LS}X_{t-1} = \epsilon_t - (\hat\phi_{LS} - 1)X_{t-1}$,

$$\hat\sigma^2 = \frac{1}{n}\sum \epsilon_t^2 - 2(\hat\phi_{LS} - 1)\cdot\frac{1}{n}\sum X_{t-1}\epsilon_t + (\hat\phi_{LS} - 1)^2 \cdot \frac{1}{n}\sum X_{t-1}^2,$$

where $\hat\phi_{LS} - 1 = O_p(n^{-1})$, $n^{-1}\sum X_{t-1}\epsilon_t = O_p(1)$ and $n^{-1}\sum X_{t-1}^2 = O_p(n)$, so the last two terms are $O_p(n^{-1})$. One rejects H_0 when $|DF_n| \geq DF_\alpha$, where $P(|DF| \geq DF_\alpha) = \alpha$.

4.2.2 The AR(p) case: augmented Dickey–Fuller test

Analogous tests exist for models with drift or trend; we treat the pure AR(p) model $\phi(B)X_t = \epsilon_t$, $\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$, and test whether $\phi(1) = 0$.

Lemma (Error-correction form)

For any $\phi_1, \ldots, \phi_p$, with $\Delta X_t = X_t - X_{t-1}$,

$$X_t - \sum_{i=1}^{p}\phi_i X_{t-i} = X_t - \beta X_{t-1} - \sum_{j=1}^{p-1}\psi_j \Delta X_{t-j}, \qquad \beta = \sum_{i=1}^{p}\phi_i = 1 - \phi(1), \quad \psi_j = -\sum_{i=j+1}^{p}\phi_i.$$

In particular $\phi(1) = 0$ iff $\beta = 1$, and then $\phi(B) = \psi(B)(1 - B)$ with $\psi(B) = 1 - \psi_1 B - \cdots - \psi_{p-1} B^{p-1}$.

Proof. [Supplement] Since $X_{t-i} = X_{t-1} - \sum_{j=1}^{i-1} \Delta X_{t-j}$ for $i \geq 1$,

$$\sum_{i=1}^{p} \phi_i X_{t-i} = \Big(\sum_{i=1}^{p} \phi_i\Big) X_{t-1} - \sum_{i=1}^{p} \phi_i \sum_{j=1}^{i-1} \Delta X_{t-j} = \beta X_{t-1} - \sum_{j=1}^{p-1} \Big(\sum_{i=j+1}^{p} \phi_i \Big) \Delta X_{t-j},$$

after exchanging the order of summation; this is the identity. If $\beta = 1$ the right-hand side equals $\Delta X_t - \sum_j \psi_j \Delta X_{t-j} = \psi(B)(1 - B) X_t$. ■

Hence the AR(p) model can be written as the regression

$$X_t = \beta X_{t-1} + \psi_1 \Delta X_{t-1} + \cdots + \psi_{p-1} \Delta X_{t-p+1} + \eta_t, \qquad \{\eta_t\} \sim \mathrm{IID}(0, \tau^2), \tag{4.4}$$

and under the unit-root hypothesis $\beta = 1$ and $\eta_t = \epsilon_t$. The least-squares estimator $\hat{\beta}$ from (4.4) should then be close to 1, and one rejects for large $\left|(\hat{\beta} - 1)/\operatorname{sd}(\hat{\beta})\right|$. This is the *augmented Dickey–Fuller* (ADF) test; the lagged differences absorb the short-run dependence, and the statistic has the same Dickey–Fuller limit law DF as in the AR(1) case.

4.3 Model selection and diagnostics

Motivation. Good forecasts require the right model. We describe the classical Box–Jenkins identification by sample autocorrelations, then information criteria (AIC, BIC) that select orders automatically, and finally checks of a fitted model by overfitting and residual analysis.

4.3.1 The Box–Jenkins method

The Box–Jenkins procedure for ARMA models has three stages:

(1) *identification* (choose p and q), (2) *estimation* of the parameters (Chapter 3), and (3) *diagnostic checking* (Section 4.3.3). If the model fails the checks one returns to stage 1 (Figure 1).

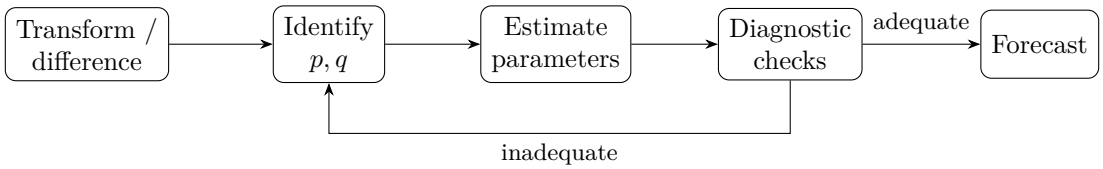


Figure 1: The Box–Jenkins modelling cycle.

Making the series stationary. If the spread of the series is not constant, a variance-stabilizing transformation is applied first: the Box–Cox transform $(y^\lambda - 1)/\lambda$ ($\lambda \neq 0$), or $\sqrt{y}$, $1/y$, $\log y$ (the log is standard for economic data). If the series trends up or down it is differenced. For a stationary series the sample ACF decays to 0 quickly, while for a nonstationary one it decays very slowly; a unit-root test (Section 4.2) settles whether differencing is needed. Usually $d \in \{0, 1, 2\}$.

Identification by ACF and PACF. After transformation, the sample ACF (SACF) and sample PACF (SPACF) are compared with the theoretical patterns of Chapter 2 (Table 1); the SACF mainly guides q and the SPACF mainly guides p. Large p, q increase the number of parameters ($p + q + 2$

Model	ACF	PACF
AR(p)	decays exponentially or as a damped sine	cuts off after lag p
MA(q)	cuts off after lag q	decays exponentially or as a damped sine
ARMA(p,q)	decays after lag $q-p$	decays after lag $p-q$

Table 1: ACF and PACF of ARMA models (Table 4.3.1).

including mean and variance) and lead to inefficient, complicated models, so parsimonious choices are preferred; usually $p \le 3$, $q \le 3$.

For an AR(p) model the PACF satisfies $\phi_{kk} = 0$ for $k \ge p+1$, so one tests $H_0 : \phi_{kk} = 0$ vs $H_1 : \phi_{kk} \neq 0$. For $k \ge p+1$, $\hat{\phi}_{kk}$ is approximately normal with $E\hat{\phi}_{kk} \approx 0$ and $\operatorname{Var}\hat{\phi}_{kk} \approx 1/n$; hence H_0 is rejected at level α when $|\hat{\phi}_{kk}| > z_{\alpha/2}/\sqrt{n}$, and $|\hat{\phi}_{kk}| \le z_{\alpha/2}/\sqrt{n}$ is read as $\phi_{kk} = 0$.

For an MA(q) model the ACF cuts off after lag q; one tests $H_0 : \rho(k) = 0$ and rejects when $|\hat{\rho}(k)| > z_{\alpha/2}\,\mathrm{S.E.}(\hat{\rho}(k))$, where by Bartlett's formula

$$\mathrm{S.E.}(\hat{\rho}(k)) \approx \Big(\frac{1 + 2\sum_{j=1}^{q}\hat{\rho}(j)^2}{n}\Big)^{1/2}, \qquad k \ge q+1.$$

If $|\hat{\rho}(k)| \le z_{\alpha/2}\,\mathrm{S.E.}(\hat{\rho}(k))$ for all $k \ge q+1$, an MA(q) model is tentatively chosen. For mixed ARMA models neither function cuts off, and the orders are chosen by the criteria of the next subsection.

4.3.2 Model selection criteria

Choosing a model means choosing a parametric family. For data $X_1, \dots, X_n$ consider, for example, the nested normal models

$$M_1 = \{N(\mu, 1) : -\infty < \mu < \infty\}, \qquad M_2 = \{N(\mu, \sigma^2) : -\infty < \mu < \infty,\ 0 < \sigma^2 < \infty\}.$$

If M_1 is correct, choosing M_2 wastes a parameter (inefficiency); if M_2 is correct, choosing M_1 ignores an essential parameter, which is worse. Akaike's approach: let $f_1(x_1, \dots, x_n; \mu)$ and $f_2(x_1, \dots, x_n; \mu, \sigma^2)$ be the joint densities, compute the maximized likelihoods $MLE_1 = f_1(X_1, \dots, X_n; \hat{\mu})$ and $MLE_2 = f_2(X_1, \dots, X_n; \hat{\mu}, \hat{\sigma}^2)$, and

$$AIC(M_1) = -2\log(MLE_1) + 2 \times 1, \qquad AIC(M_2) = -2\log(MLE_2) + 2 \times 2;$$

the model with the smaller value is selected. In general

$$AIC(\text{model}) = -2\log(MLE) + 2\,(\text{number of parameters in the model}), \tag{4.5}$$

$$BIC(\text{model}) = -2\log(MLE) + \log(n)\,(\text{number of parameters in the model}).$$

The maximized likelihood always increases with the number of parameters, so it alone would favour ever larger models, contradicting parsimony; the second term penalizes the number of parameters and offsets this. The factor 2 in front of the log-likelihood is conventional. Unlike Box–Jenkins identification, information criteria apply directly to ARMA models and rest on a complete theory (AIC estimates the expected Kullback–Leibler discrepancy between the fitted and the true model).

[Supplement] AIC versus BIC

Compare a correct model M with an overfitted model $M' \supset M$ having one more parameter. The likelihood-ratio statistic $\Lambda_n = 2\log(MLE') - 2\log(MLE)$ is asymptotically χ^2_1 under M. AIC selects M' iff $\Lambda_n > 2$, which has limiting probability $P(\chi^2_1 > 2) \approx 0.157 > 0$; BIC selects M' iff $\Lambda_n > \log n$, whose probability tends to 0. If instead M is false and M' true, Λ_n/n

converges to a positive constant (a Kullback–Leibler divergence), so both criteria reject M with probability tending to one. Hence BIC is consistent for the true order, while AIC tends to overfit (but is efficient for prediction).

4.3.3 Model diagnostics

After identification and estimation, *diagnostic checking* asks whether the tentative model is adequate. If it is, it becomes the forecasting model; otherwise one returns to identification. Two tools are used: overfitting and residual analysis.

(1) Overfitting. If ARIMA(p,d,q) was chosen, also fit the larger models ARIMA$(p+1,d,q)$ and ARIMA$(p,d,q+1)$ and ask:

(a) are the added parameters judged to be zero?

(b) are the estimates of the original parameters essentially unchanged?

If both answers are yes, the tentative model passes; otherwise another model should be tried. For instance, if MA(1) was chosen, fit ARMA(1,1) and MA(2) and check that ϕ_1 and θ_2 are insignificant and that the three estimates of θ_1 agree. The added estimates are asymptotically normal with mean 0 under H_0: "the tentative model is true", which yields the tests.

(2) Residual analysis. Suppose the selected model is the stationary ARMA(p,q) model

$$\phi(B)(X_t-\mu)=\theta(B)\epsilon_t. \tag{4.6}$$

If it is correct and well estimated, the residuals $\{e_t\}$ should behave like the IID errors $\{\epsilon_t\}$. With estimates $\hat\phi_1,\dots,\hat\phi_p,\hat\theta_1,\dots,\hat\theta_q,\hat\mu$ and $\hat\phi(B)=1-\hat\phi_1B-\cdots-\hat\phi_pB^p$, $\hat\theta(B)=1+\hat\theta_1B+\cdots+\hat\theta_qB^q$, the residuals are

$$e_t=\hat\theta(B)^{-1}\hat\phi(B)(X_t-\hat\mu),\qquad t=1,\dots,n,$$

computed with $X_t=0$ for $t\le 0$. One checks:

(a) is the mean zero? (b) is the variance constant? (c) are the residuals independent, i.e. are their autocorrelations negligible? (a) is checked by comparing $\bar e$ with its standard error, (b) by a residual plot. For (c) use the residual sample ACF

$$\rho_e(k)=\frac{\sum_{t=1}^{n-k}(e_t-\bar e)(e_{t+k}-\bar e)}{\sum_{t=1}^{n}(e_t-\bar e)^2}.$$

Under H_0: "(4.6) is the true model" (so $\rho(k)=0$ for $k\ge1$), $\mathrm{Var}(\rho_e(k))\approx 1/n$, and the approximate rejection region at level α is $|\rho_e(k)|\ge z_{\alpha/2}\,\mathrm{S.E.}(\rho_e(k))$. One may also fit ARMA models to the residuals themselves: ARMA(0,0) (white noise) should be identified.

Testing each lag separately at level α ignores the dependence among the $\rho_e(k)$ and inflates the overall type I error, whereas the real hypothesis is the simultaneous one that all residual autocorrelations vanish. The *portmanteau* or *Ljung–Box* statistic

$$Q=n(n+2)\sum_{h=1}^{K}\frac{\rho_e^2(h)}{n-h}$$

is approximately $\chi^2_{K-p-q-1}$ under H_0; the model is rejected at level α if Q exceeds the upper $100\alpha\%$ point of $\chi^2_{K-p-q-1}$.

The *Durbin–Watson* statistic tests no first-order correlation against AR(1) errors:

$$DW=\frac{\sum_{t=2}^{n}(e_t-e_{t-1})^2}{\sum_{t=1}^{n}e_t^2}\approx 2(1-\rho_e(1)),$$

since the numerator equals $\sum_{t\geq 2} e_t^2 + \sum_{t\geq 2} e_{t-1}^2 - 2\sum_{t\geq 2} e_t e_{t-1} \approx 2\sum e_t^2 - 2\sum e_t e_{t-1}$. Thus $0 \leq DW \leq 4$ approximately; $DW \approx 2$ means no lag-one correlation, $DW < 2$ positive and $DW > 2$ negative correlation. Critical values are tabulated by Durbin and Watson (1971).

4.4 Selected exercises

Motivation. These textbook exercises make the differencing argument of Section 4.1 rigorous, treat stochastic trends and over-differencing, and establish the basic properties of the random walk and of Brownian motion used in the unit-root theory. The algebraic identities for T_n already derived in Section 4.2.1 are not repeated.

Exercise 4.1 (Differencing a polynomial trend)

Let $X_t = a_0 + a_1 t + \cdots + a_r t^r + \epsilon_t$ with $r \geq 1$, $a_r \neq 0$, $\{\epsilon_t\} \sim WN(0, \sigma^2)$. Show that
(a) $(1-B)^r t^r = r!$;
(b) $(1-B)^r \epsilon_t = \sum_{j=0}^{r} \binom{r}{j} (-1)^j \epsilon_{t-j}$;
(c) $(1-B)^r X_t$ is stationary, and find its mean and variance.

Solution.

(a)

Step 1. Let p be a polynomial of degree $k \geq 1$ with leading coefficient c. By the binomial theorem,

$$(1-B)t^k = t^k - (t-1)^k = -\sum_{i=0}^{k-1} \binom{k}{i} (-1)^{k-i} t^i = k\, t^{k-1} + (\text{terms of degree} \leq k-2),$$

so by linearity $(1-B)p$ has degree $k-1$ and leading coefficient kc. For a constant c, $(1-B)c = c - c = 0$.

Step 2. By induction on $j = 0, \ldots, r$, $(1-B)^j t^r$ is a polynomial of degree $r-j$ with leading coefficient $r(r-1)\cdots(r-j+1)$. For $j = r$ this is the constant $r!$.

(b) The operators 1 and B commute, so the binomial theorem holds for $(1-B)^r$: $(1-B)^r = \sum_{j=0}^{r} \binom{r}{j} (-1)^j B^j$. Applying it to ϵ_t and using $B^j \epsilon_t = \epsilon_{t-j}$ gives the claim.

(c)

Step 1. For $k < r$, Step 2 of (a) with $j = k$ gives $(1-B)^k t^k = k!$, a constant, so $(1-B)^r t^k = (1-B)^{r-k} k! = 0$ as $r - k \geq 1$. By linearity and (a),(b),

$$Z_t := (1-B)^r X_t = a_r\, r! + \sum_{j=0}^{r} c_j \epsilon_{t-j}, \qquad c_j = \binom{r}{j} (-1)^j.$$

Step 2. $Z_t - a_r r!$ is a finite moving average of white noise. Its ACVF is computed exactly as in (2.15) (only $\mathrm{Cov}(\epsilon_s, \epsilon_t) = \sigma^2 \mathbb{1}\{s = t\}$ is used): $\gamma_Z(h) = \sigma^2 \sum_{j=0}^{r-|h|} c_j c_{j+|h|}$ for $|h| \leq r$ and 0 otherwise, free of t. Hence $\{Z_t\}$ is stationary with mean $EZ_t = a_r\, r!$.

Step 3. By Vandermonde's identity $\sum_j \binom{r}{j}\binom{r}{r-j} = \binom{2r}{r}$,

$$\mathrm{Var}(Z_t) = \sigma^2 \sum_{j=0}^{r} \binom{r}{j}^2 = \binom{2r}{r} \sigma^2.$$

■

Exercise 4.2 (Markov property of the random walk)

Let $X_t = X_{t-1} + \epsilon_t$, $X_0 = 0$, $\{\epsilon_t\} \sim IID(0, \sigma^2)$ with values in a countable set. Show that, whenever the conditioning events have positive probability,

$$P(X_n = i_n \mid X_0 = i_0, X_1 = i_1, \ldots, X_{n-1} = i_{n-1}) = P(X_n = i_n \mid X_{n-1} = i_{n-1}).$$

Solution.

Step 1. $X_k = \epsilon_1 + \cdots + \epsilon_k$, so $X_0, \ldots, X_{n-1}$ are functions of $\epsilon_1, \ldots, \epsilon_{n-1}$ and are independent of ϵ_n. Let $A = \{X_0 = i_0, \ldots, X_{n-1} = i_{n-1}\}$ and $A' = \{X_{n-1} = i_{n-1}\}$; both belong to $\sigma(\epsilon_1, \ldots, \epsilon_{n-1})$.

Step 2. On A we have $X_n = i_{n-1} + \epsilon_n$, so by independence

$$P(X_n = i_n, A) = P(\epsilon_n = i_n - i_{n-1}, A) = P(\epsilon_n = i_n - i_{n-1})P(A).$$

Dividing by $P(A) > 0$: $P(X_n = i_n \mid A) = P(\epsilon_1 = i_n - i_{n-1})$, using that the ϵ_t are identically distributed.

Step 3. The same computation with A' in place of A gives $P(X_n = i_n \mid A') = P(\epsilon_1 = i_n - i_{n-1})$. The two conditional probabilities coincide.

For general (not necessarily discrete) ϵ_t with distribution function F, the same argument with the independence ("freezing") lemma gives $P(X_n \le x \mid X_0, \ldots, X_{n-1}) = F(x - X_{n-1})$, which is a function of X_{n-1} alone and therefore also equals $P(X_n \le x \mid X_{n-1})$. ■

Exercise 4.3 (Quadratic trend plus a local linear trend)

Let

$$X_t = \alpha_0 + \alpha_1 t + \alpha_2 t^2 + \mu_t + \epsilon_t, \qquad \mu_t = \mu_{t-1} + \beta_t + \eta_t, \qquad \beta_t = \beta_{t-1} + \zeta_t,$$

where $\{\epsilon_t\} \sim WN(0, \sigma_\epsilon^2)$, $\{\eta_t\} \sim WN(0, \sigma_\eta^2)$, $\{\zeta_t\} \sim WN(0, \sigma_\zeta^2)$ are mutually independent. Difference $\{X_t\}$ to obtain a stationary series and find its mean and variance.

Solution.

Step 1. From the recursions, $(1-B)\beta_t = \zeta_t$ and $(1-B)\mu_t = \beta_t + \eta_t$, hence

$$(1-B)^2\mu_t = (1-B)\beta_t + (1-B)\eta_t = \zeta_t + \eta_t - \eta_{t-1}.$$

Step 2. By Exercise 4.1(a) and Step 1 of its part (c), $(1-B)^2(\alpha_0 + \alpha_1 t + \alpha_2 t^2) = 2\alpha_2$, and $(1-B)^2\epsilon_t = \epsilon_t - 2\epsilon_{t-1} + \epsilon_{t-2}$. Therefore

$$D_t := (1-B)^2 X_t = 2\alpha_2 + \zeta_t + (\eta_t - \eta_{t-1}) + (\epsilon_t - 2\epsilon_{t-1} + \epsilon_{t-2}).$$

Step 3. The three noise terms are uncorrelated with each other, and each is a finite moving average of white noise with ACVF given by (2.15) (with coefficients (1), $(1, -1)$, $(1, -2, 1)$). By Exercise 1.7 (which only uses zero cross-covariances), $\{D_t\}$ is stationary with

$$\gamma_D(0) = \sigma_\zeta^2 + 2\sigma_\eta^2 + 6\sigma_\epsilon^2, \quad \gamma_D(\pm 1) = -\sigma_\eta^2 - 4\sigma_\epsilon^2, \quad \gamma_D(\pm 2) = \sigma_\epsilon^2,$$

and $\gamma_D(h) = 0$ for $|h| \ge 3$.

Step 4. Hence $ED_t = 2\alpha_2$ and $\mathrm{Var}(D_t) = \sigma_\zeta^2 + 2\sigma_\eta^2 + 6\sigma_\epsilon^2$.

One difference does not suffice in general: $(1-B)X_t = \alpha_1 - \alpha_2 + 2\alpha_2 t + \beta_t + \eta_t + \epsilon_t - \epsilon_{t-1}$ has a mean linear in t when $\alpha_2 \neq 0$, and contains the random walk β_t, whose variance grows linearly in t when $\sigma_\zeta^2 > 0$. ■

Exercise 4.8 (Properties of Brownian motion)

Let $\{W(s) : s \geq 0\}$ be a standard Brownian motion (Definition 4.2.1). Show that
(a) $EW(t)W(s) = t \wedge s$;
(b) $Z(s) = W(\eta s)/\sqrt{\eta}$ is a Brownian motion for every $\eta > 0$;
(c) $G(s) = sW(1/s)$ for $s > 0$, $G(0) = 0$, is a Brownian motion.

Solution.

(a) Let $s \leq t$. By (3) of Definition 4.2.1, $W(s)$ and $W(t) - W(s)$ are independent with mean 0, and by (1),(2), $EW(s)^2 = s$. Hence $EW(t)W(s) = EW(s)^2 + E[W(s)(W(t) - W(s))] = s = s \wedge t$.

A characterization. Before (b),(c) we prove: *a process $\{G(s)\}$ with continuous paths, $G(0) = 0$, whose finite-dimensional laws are jointly normal with mean 0 and $\mathrm{Cov}(G(s), G(t)) = s \wedge t$, is a Brownian motion; and conversely W has these properties.*

Step 1. For $t > s$, $G(t) - G(s)$ is normal with mean 0 and variance $t - 2(s \wedge t) + s = t - s$.

Step 2. For $0 \leq t_0 < t_1 < \cdots < t_k$, the increments $\Delta_i = G(t_i) - G(t_{i-1})$ are a linear image of a normal vector, hence jointly normal. For $i < j$, $t_i \leq t_{j-1}$, so

$$\mathrm{Cov}(\Delta_i, \Delta_j) = t_i \wedge t_j - t_i \wedge t_{j-1} - t_{i-1} \wedge t_j + t_{i-1} \wedge t_{j-1} = t_i - t_i - t_{i-1} + t_{i-1} = 0.$$

Jointly normal and uncorrelated variables are independent (the characteristic function factorizes, as in Exercise 1.6). With $t_0 = 0$ this gives (3) of Definition 4.2.1.

Step 3. Conversely, $(W(t_1), \ldots, W(t_k))$ is a linear image (partial sums) of the independent normal increments, hence jointly normal with mean 0, and its covariance is $s \wedge t$ by (a).

(b) Z has continuous paths and $Z(0) = 0$; its finite-dimensional laws are linear images of those of W, hence jointly normal with mean 0; and $\mathrm{Cov}(Z(s), Z(t)) = (\eta s \wedge \eta t)/\eta = s \wedge t$. The characterization applies.

(c)

Step 1. For $s, t > 0$, $\mathrm{Cov}(G(s), G(t)) = st\,(1/s \wedge 1/t) = st/(s \vee t) = s \wedge t$, and this also holds trivially if s or t is 0. The finite-dimensional laws are linear images of those of W (with the constant $G(0) = 0$), hence jointly normal with mean 0. Paths are continuous on $(0, \infty)$ as compositions of continuous maps.

Step 2. It remains to show continuity at 0, i.e. $W(u)/u \to 0$ a.s. as $u \to \infty$. Since $W(n) = \sum_{k=1}^{n}(W(k) - W(k-1))$ is a sum of i.i.d. $N(0,1)$ variables, the strong law gives $W(n)/n \to 0$ a.s.

Step 3. Let $M_k = \sup_{u \in [k, k+1]} |W(u) - W(k)|$, a supremum over rationals by path continuity. The process $u \mapsto W(k+u) - W(k)$, $u \in [0,1]$, is Gaussian with mean 0 and covariance $u \wedge v$ (by (a)), so it has the same finite-dimensional laws as W on $[0,1]$, and M_k has the law of $M_0 = \sup_{u \leq 1} |W(u)|$. By Doob's L^2 maximal inequality for the martingale W (used as a tool), $EM_0^2 \leq 4EW(1)^2 = 4$, so $EM_0 \leq 2$.

Step 4. For $\varepsilon > 0$, $\sum_{k \geq 1} P(M_k > \varepsilon k) = \sum_{k \geq 1} P(M_0/\varepsilon > k) \leq E(M_0)/\varepsilon < \infty$. By Borel–Cantelli, $M_k \leq \varepsilon k$ eventually a.s.; letting $\varepsilon \downarrow 0$ along a sequence, $M_k/k \to 0$ a.s.

Step 5. For $u \in [k, k+1]$, $|W(u)|/u \leq (|W(k)| + M_k)/k \to 0$ a.s. by Steps 2 and 4. Hence $G(s) = W(1/s)/(1/s) \to 0 = G(0)$ as $s \downarrow 0$, and the characterization shows that G is a Brownian motion. ■

Exercise 4.9 (Over-differencing an MA(2))

Let $X_t = \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2}$, $\{\epsilon_t\} \sim IID\, N(0, \sigma^2)$, be stationary, and suppose one fits $(1 - B)X_t = Y_t$. Find the ACVF of $\{Y_t\}$.

Solution.

Step 1. $Y_t = X_t - X_{t-1} = \epsilon_t + (\theta_1 - 1)\epsilon_{t-1} + (\theta_2 - \theta_1)\epsilon_{t-2} - \theta_2\epsilon_{t-3}$, an MA(3) with coefficients $c_0 = 1$, $c_1 = \theta_1 - 1$, $c_2 = \theta_2 - \theta_1$, $c_3 = -\theta_2$.

Step 2. By Exercise 2.9(a) with $c = 1$, $\gamma_Y(h) = 2\gamma_X(h) - \gamma_X(h-1) - \gamma_X(h+1)$, where by (2.15) $\gamma_X(0) = (1+\theta_1^2+\theta_2^2)\sigma^2$, $\gamma_X(\pm1) = (\theta_1+\theta_1\theta_2)\sigma^2$, $\gamma_X(\pm2) = \theta_2\sigma^2$ and $\gamma_X(h) = 0$ for $|h| \geq 3$. Hence

$$\begin{aligned}
\gamma_Y(0) &= 2\sigma^2(1 + \theta_1^2 + \theta_2^2 - \theta_1 - \theta_1\theta_2),\\
\gamma_Y(\pm1) &= \sigma^2(2\theta_1 + 2\theta_1\theta_2 - \theta_2 - 1 - \theta_1^2 - \theta_2^2),\\
\gamma_Y(\pm2) &= \sigma^2(2\theta_2 - \theta_1 - \theta_1\theta_2), \qquad \gamma_Y(\pm3) = -\theta_2\sigma^2,
\end{aligned}$$

and $\gamma_Y(h) = 0$ for $|h| \geq 4$. (These agree with $\sigma^2 \sum_j c_j c_{j+|h|}$ from Step 1.)

Step 3. The MA polynomial of Y is $(1-z)\theta(z)$, which vanishes at $z = 1$: the over-differenced series is not invertible. For example, $\theta_1 = \theta_2 = 0$ gives $\rho_Y(1) = -\frac{1}{2}$; a lag-one sample autocorrelation near -0.5 is a warning sign of unnecessary differencing. ■